

Project Plan - Agents with Small Language Models

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Motivation

Recent progress in agentic systems has primarily resulted from the development of increasingly large language models [1], whose scale facilitates intricate reasoning, tool usage, and structured interactions. However, reliance on large models poses major challenges, including elevated computing costs, elevated latency, greater deployment complexity, and restricted accessibility in resource-constrained settings. This situation motivates an investigation into whether current agent capabilities are inherently tied to model scale, or whether comparable reliability and functionality can be achieved through enhanced system design using smaller language models [2] and [3].

This thesis is motivated by the hypothesis that many shortcomings observed in agentic systems, including unreliable function calls, malformed structured outputs, and nondeterministic behavior, emerge primarily from insufficient constraints during generation rather than from model size. Constrained decoding and structural enforcement techniques deliver an effective alternative, enabling smaller language models to generate outputs that strictly adhere to fixed schemas and execution plans [4]. Guiding generation through formal constraints may enable predictable, production-ready agent behavior without reliance on ever-larger models.

This research aims to determine whether techniques currently employed in large-model agentic systems can be effectively replicated using small language models when supported by constrained decoding. Demonstrating this capability would help decouple agent reliability from model scale, thereby enabling the development of lightweight, cost-efficient, and locally deployable agents suitable for enterprise, embedded, and privacy-sensitive applications.

Ultimately, this research seeks to contribute to a shift in focus from scaling models to improving system-level reliability, demonstrating that dependable agentic systems can be built through structural and algorithmic improvements rather than computational scale alone.

Background

One of the main challenges for LLM-based agents is reliably generating structured outputs, especially when calling functions or tools. These models need to follow specific schemas, using valid function names and correct argument types. However, when generation is not controlled, outputs can be malformed, may contain missing fields, or may include tools that do not exist. This often means more post-processing and repeated queries, which makes the system less efficient and reliable.

To solve these problems, some research has explored constrained decoding [5] and structural enforcement. This makes language generation a controlled process, improving consistency and accuracy in agent workflows. For function and tool calls, constrained decoding lets models choose

only from registered functions and create arguments that meet the required specifications, helping avoid common errors in small and medium-sized models.

Also, several methods have been developed to make small language models more efficient, including knowledge distillation [6], low-rank factorization [7], quantization [8], and pruning [9]. These techniques reduce the need for computing power and memory, enabling the models to run on devices with limited resources. However, these methods do not directly improve output reliability, and smaller or compressed models can still produce invalid outputs if decoding is not controlled. Methods and decoding-level control are largely complementary. Integrating lightweight models with structural enforcement mechanisms provides a pathway to systems that are both computationally efficient and behaviorally reliable.

In this context, the thesis points to constrained decoding for function and tool calling as a key way to achieve reliable behavior in small language models. By studying structural enforcement alongside efficiency methods, this research examines how careful system design can compensate for limited model capacity and help build controllable, efficient agent systems.

Research Question

Can reliable agentic behavior, particularly in function calling and structured output generation, be achieved using small language models when constrained decoding techniques are applied, thereby reducing dependence on large-scale models for production-grade agent systems?

This research is based on the assumption that many reliability issues in current agentic systems arise not only from limited model capability but also from unconstrained generation processes that permit outputs to deviate from expected structures. The central hypothesis is that enforcing structural constraints during decoding enables small language models to achieve levels of correctness, determinism, and robustness comparable to those of larger models in agent scenarios requiring precise tool invocation and structured responses.

Addressing this question provides new insight into whether advances in agentic AI primarily depend on model scale or on improvements in generation control and system design. Demonstrating that constrained decoding can enable dependable agent behavior in small models would support the development of more efficient, cost-effective, and deployable AI systems, particularly in environments with limited computational resources or where local deployment is necessary.

Time plan

The main phases of this thesis are outlined below. An overview of the timetable associated with these phases is provided in Fig. 1.

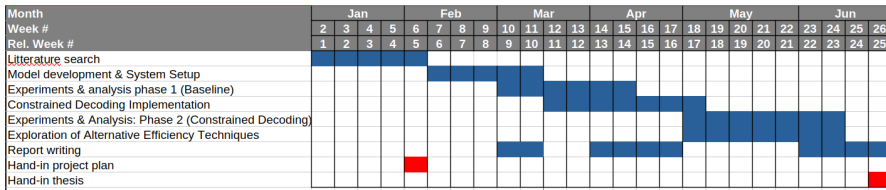


Figure 1: Gantt chart of the project plan.

Several risks may impact the project timeline. The literature review phase could extend beyond initial estimates if relevant prior work is broader or more fragmented than anticipated,

potentially delaying subsequent design decisions. Model development and system setup may encounter technical challenges related to framework compatibility, hardware limitations, or unforeseen implementation complexity.

During the experimental phases, delays may arise from unstable model behavior, longer-than-expected training or inference times, or the need to repeat experiments to ensure statistical validity. Exploring alternative efficiency techniques, such as knowledge distillation or model compression, carries the risk of limited performance gains or increased engineering overhead, which may reduce the depth of analysis achievable within the allocated timeframe. To mitigate these risks, the project prioritizes constrained decoding as the core contribution and maintains flexibility in the scope of complementary techniques.

Finally, report writing may be affected by delays in earlier phases. To address this, writing is scheduled to overlap with experimentation to ensure steady progress toward the final submission.

References

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